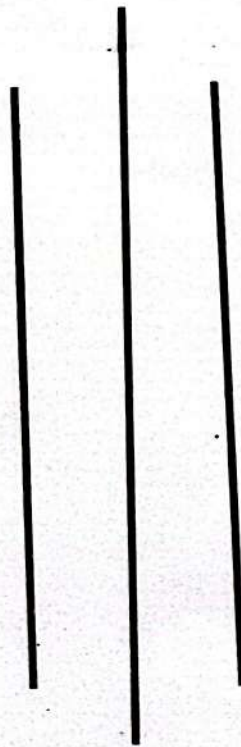


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Lab report on
Data Communication and Networks [ICT Ed. 456]

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TITLE : INSTALLATION OF WIRESHARK AND LEARNING TO USE IT.

OBJECTIVES:

- To install Wireshark on the system.
- To understand basic packet capturing concepts.
- To observe network traffic using Wireshark.

TOOLS USED:

- Wireshark
- Npcap

PROBLEM ANALYSIS :

In computer networks, data communication takes place through packets, but these packets are not directly visible to users. Without proper tools, it becomes difficult to understand how data flows across a network.

Wireshark is a free packet sniffing tool that allows users to capture and analyze network packets in real time. By installing and using Wireshark, users can gain practical knowledge of network protocols and traffic flow.

STEPS: PERFORMED:

- i) Visited the official website of Wireshark and download the installer.
- ii) Installed Wireshark & selected the option to install Npcap.
- iii) Launched Wireshark and viewed available network interfaces.
- iv) Selected the wifi interfaces and started live capture.
- v) Opened a website to generate network traffic.
- vi) Stopped the capture after packets appeared.
- vii) Applied basic filters such as http and ip addr.
- viii) Viewed packet layers in Packet list, Details and Bytes panes.

OUTPUT:

The output arrives automatically because Wireshark captures the packets your computer is sending/receiving when you browse the Internet.

DISCUSSION AND CONCLUSION:

This lab focused on introducing Wireshark as a packet analysis tool and understanding its role in monitoring network traffic. Installing Wireshark helps learners visualize real-time data transmission and understand how different protocols operate within a network.

Michael Hayes

TITLE: CAPTURE AND SAVE NETWORK TRAFFIC USING WIRESHARK.

OBJECTIVES:

- To capture live network packets using Wireshark.
- To generate and observe network traffic.
- To save capture data in .pcapng format for analysis.

TOOLS USED:

- Wireshark
- Npcap

PROBLEM ANALYSIS:

In computer networks, data is transmitted in the form of packets, which are not visible without special tools. This makes network analysis difficult. Wireshark helps capture and monitor live traffic. In this lab, packets are captured and saved to understand network communication.

STEPS:

- i) Open Wireshark on your system.
- ii) Select the active network interface (Wifi / Ethernet).
- iii) Start capturing live network traffic.
- iv) Generate traffic by opening a website or using the ~~ping~~ command.
- v) ~~Capture~~ for 2-3 minutes and stop the capture.
- vi) Save the captured file in .pcapng format.

OUTPUT:

- Active network interface was selected successfully.
- Live network traffic was captured for 2-3 minutes.
- Traffic was generated using web browsing or ping command.
- Different protocols such as TCP, HTTP and DNS were observed.
- The captured file was saved in .pcapng format.

DISCUSSION AND CONCLUSION:

This lab showed how to capture and save network traffic using Wireshark. It helped in understanding how data flows in a network. The experiment provides basic knowledge useful for network analysis and troubleshooting.

Nikhil

TITLE: SUBNETTING AND NETWORK COMMUNICATION ANALYSIS USING WIRESHARK.

OBJECTIVES:

- To divide a network into smaller subnets and assign IP addresses.
- To test device communication using ping.
- To capture and analyze packets using Wireshark.

TOOLS USED:

- Wireshark

PROBLEM ANALYSIS:

Dividing a network into subnets improves management and efficiency. Understanding subnetting and analyzing communication between devices is essential for network design. This lab helps visualize IP assignment, ping testing and packet capture in smaller networks.

STEPS:

- i) Take network 192.168.10.0/24 and divide into 4 subnets.
- ii) Calculate subnet mask, subnet IDs, broadcast addresses, and host ranges.
- iii) Assign IP addresses to at least two devices in one subnet.
- iv) Test communication using ping command.
- v) Capture packets using Wireshark.
- vi) Identify ICMP packets in the capture.

OUTPUT:

Subnet Table:

Subnet	Subnet ID	Broadcast	Host Range
1	192.168.10.0	192.168.10.63	192.168.10.1-62
2	192.168.10.64	192.168.10.127	192.168.10.65-126
3	192.168.10.128	192.168.10.191	192.168.10.129-190
4	192.168.10.192	192.168.10.255	192.168.10.193-254

IP Assignment Example:

- PC1 → 192.168.10.1/26
- PC2 → 192.168.10.2/26

Ping Test: Success between assigned devices.

Packet Capture: ICMP packets identified in Wireshark

DISCUSSION AND CONCLUSION:

This lab demonstrated subnetting and device communication in a network. By assigning IPs, testing ping, and analyzing captured packets, the flow of data between device was observed. The experiment helps in understanding subnetting, packet structure, and basic network troubleshooting.

Richard Gray

TITLE: ANALYSIS OF TCP AND UDP TRAFFIC USING WIRESHARK.

OBJECTIVES:

- To capture and analyze UDP and TCP packets using Wireshark.

TOOLS USED:

- Wireshark

PROBLEM ANALYSIS:

TCP and UDP are important transport layer protocols used in network communication. TCP is connection-oriented and ensures reliable data transfer, while UDP is connectionless and faster but less reliable.

STEPS:

For UDP:

- i) Open Wireshark on your system.
- ii) Select the active network interface (Wifi).
- iii) Start capturing packets.
- iv) Generate UDP traffic by opening a website.
- v) Apply filters udp or dns in the filter bar.
- vi) Select any UDP packet.
- vii) Observe the UDP header fields such as Source Port, Destination Port, Length, and Checksum.

For TCP:

- i) Restart the packet capture.
- ii) Open a website using a web browser to generate TCP traffic.
- iii) Apply filter tcp in the filter bar.
- iv) Use filter tcp.flags.syn == 1 && tcp.flags.ack == 0 to find the SYN packet.

- v> Locate the corresponding SYN-ACK packet.
- v> Identify the final ACK packet to complete the three-way handshake.
- v> Stop the capture after observation.

OUTPUT:

UDP Analysis:

- UDP packets were successfully captured.
- Source Port and Destination Port were identified.
- Length & Checksum fields were observed in the UDP header.

TCP Analysis:

- TCP packets were captured during web browsing.
- Three way handshake was identified.
 - SYN (client request)
 - SYN-ACK (server response)
 - ACK (connection established).

DISCUSSION AND CONCLUSION:

This lab demonstrated the difference between TCP and UDP protocols through packet analysis. UDP packets were observed to be simple and connectionless, while TCP showed a reliable connection setup using a three-way handshake.

TITLE: ANALYSIS OF ETHERNET AND IEEE 802.11 FRAMES USING WIRESHARK.

OBJECTIVES:

- To capture and analyze Ethernet and Wi-Fi frames using Wireshark.
- To understand the structure of Ethernet and 802.11 frames.

TOOLS USED:

- Wireshark

PROBLEM ANALYSIS:

Data in networks is transmitted in the form of frames at the data link layer. Ethernet and Wi-Fi use different frame formats and addressing methods.

STEPS:

For Ethernet:

- i) Open Wireshark and select the active network interface.
- ii) Start capturing packets.
- iii) Apply filter eth to view Ethernet frames.
- iv) Select any Ethernet frame.
- v) Expand Ethernet II in the details pane.
- vi) Observe Destination MAC, source MAC, Type/Length, payload, and FCS.

For 802.11 (Wi-Fi):

- i) Ensure Wi-Fi interface is selected.
- ii) Start capturing packets.
- iii) Apply filter wlan to view Wi-Fi frames.
- iv) Select any 802.11 frame.
- v) Expand IEEE 802.11 in the details pane.
- vi) Observe Frame Control, duration, Receiver/Transmitter addresses, and Sequence Control.
- vii) Identify frame type (Management, Control, or Data).

OUTPUT:

For Ethernet:

- Ethernet frames were successfully captured and analyzed.
- Source and Destination MAC addresses along with Type/Length field were identified.
- Payload data was observed and MAC addressing role was understood.

For 802.11 (Wi-Fi):

- Wi-Fi (802.11) frames were successfully captured.
- Frame Control, Duration, and address fields were identified.
- Different frame types were observed and analyzed.

DISCUSSION AND CONCLUSIONS:

This lab demonstrated the structure and working of Ethernet and Wi-Fi frames. The experiment helps in understanding data link layer communication, which is important for network analysis & troubleshooting.

TITLE: ANALYSIS OF DHCP TRAFFIC USING WIRESHARK.

OBJECTIVES:

- To capture and analyze DHCP packets using Wireshark.
- To understand automatic IP configuration using DHCP.

TOOLS USED:

- Wireshark

PROBLEM ANALYSIS:

When a device connects to a network, it needs an IP address to communicate. Manually assigning IPs is difficult in large networks. DHCP automates this process by dynamically assigning IP addresses.

STEPS:

For DHCP Capture:

- i) Open Wireshark and select the active network interface.
- ii) Start capturing packets.
- iii) Disconnect & reconnect to the network to generate DHCP traffic.
- iv) Apply filter dhcp or bootp.
- v) Select a DHCP packet from the list.

For DHCP Analysis:

- i) Expand the DHCP section in packet details.
- ii) Identify Client MAC Address.
- iii) Identify Assigned IP Address.
- iv) Identify DHCP Server Identifier.
- v) Observe Lease Time field.
- vi) Stop the capture after analysis.

OUTPUT:

- DHCP packets were successfully captured and analyzed.
- Client MAC Address, Assigned IP, DHCP server, and Lease Time were identified.
- Automatic IP assignment process using DHCP was clearly observed.

DISCUSSION AND CONCLUSION:

This lab demonstrated how DHCP automatically assigns IP addresses to devices in a network. By analyzing DHCP packets, important fields and their roles were understood.